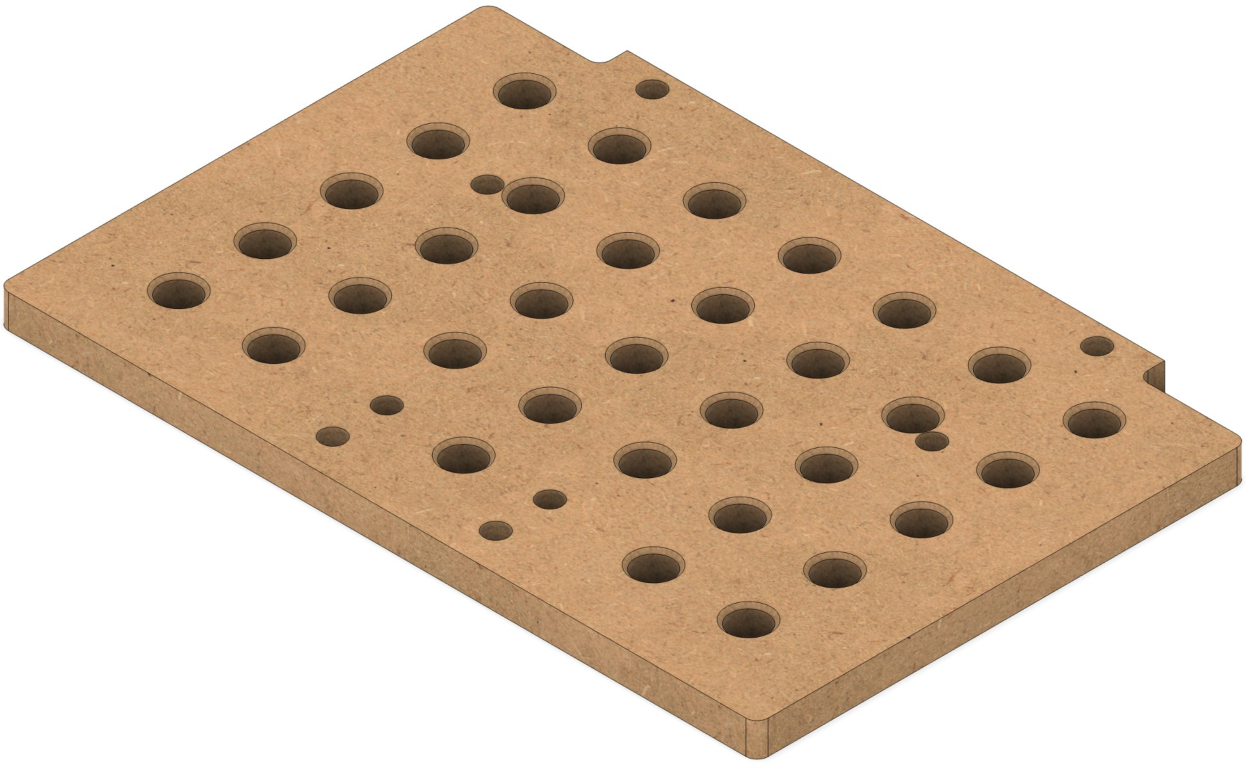
Shaper Origin Cut Instructions

Shelf Top Variations for Large Shelf



This document provides guidance on how to use the three cut files (SVG) for Shaper Origin. Each file has colour coded lines and this document explain what bit and cutting operation to use.

Design Variations

There are three variations of the Shaper bench-top large shelf design here. They are

1. Standard shelf top design
2. MFT shelf top design
3. Vacudog shelf top design

What's in each file

File	Holes found	What it is
shelfTopMFT	66× 20mm dog holes, 24× 12.7mm holes, 12× 6.5mm holes	Full MFT-style dog hole grid
shelfTopTemplate	24× 12.7mm holes, 12× 6.5mm holes (no dog holes)	Drilling/routing template only
shelfTopVacudog	1× 20mm vacuum port (centre), 24× 12.7mm, 12× 6.5mm	Template with vacuum dog hole

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Choosing the Right MDF for Your VACUDOG Worktop

The material you choose for your worktop makes a significant difference in how well your VACUDOG system performs. Cabinet-grade or furniture-grade MDF is strongly recommended over standard construction-grade MDF. Higher density MDF (typically 700–800 kg/m^3) has a tighter fibre structure that limits unwanted air flow through the worktop, allowing your vacuum pump to build the pressure differential needed for strong, reliable work-holding. Low-density or builder-grade MDF is too porous — air bleeds through it too freely, and your pump will struggle to maintain adequate suction, especially as you cut more kerfs into the surface over time.

If Cabinet-Grade MDF Is Not Available

If you can only source standard or lower-grade MDF, you can seal the faces and edges with **Minwax® Wipe-On Polyurethane, Clear Satin (946 mL/1 qt)**. This oil-based polyurethane penetrates into the MDF surface and, once cured, significantly reduces air permeability. Apply two to three thin coats, allowing full drying time between coats, sanding lightly with 320-grit between applications. Pay particular attention to sealing all four edges of the worktop, as edge grain on MDF is highly porous and a common source of vacuum leaks. Note that sealing the face of the worktop this way is appropriate for the non-cutting surface (the underside or the areas around your dog hole grid) — you want the surface your workpiece sits on to remain flat and smooth, not built up with finish layers that could affect coplanarity.

Fabrication Setup Recommendations

1. Lay your rough MDF blank on a sacrificial spoil-board on your bench
2. Have enough space on the MDF for 16.14” (410mm) by 11.02” (280 mm)
3. Apply ShaperTape in a grid pattern across the face of the MDF — keep strips at least ~60mm (2.36”) from your cut lines so the Origin can always see tape while routing near edges
4. Use 2 sided tape under the MDF to ensure the stock does not move
5. Clamp the blank (and spoil-board underneath) firmly to your bench
6. Scan, place your design, cut holes first, profile last

Order of operations (why this sequence matters)

Always cut the interior features — corner reliefs, then the small holes, then the dog holes — before you cut the board profile. While the profile is uncut, the workpiece is still a rectangular blank that sits flat and stable on the Workstation shelf with no risk of shifting. Once you cut the profile, the offcuts fall away and the part can move. See figure 1 for full cutting details. Note that I will use the 6mm cutter for all cuts except the outside cut, which will be done with a 1/4” bit.

- **Black outline** → Outside cut using 6mm bit (6mm per pass)
- **Ø20 centre circle** → Inside cut, full depth (19mm) using 6mm bit (6mm per pass)
- **Ø12.7 circles** → Pocket, 12mm deep using 6mm bit (6mm per pass)
- **Ø6.5 circles** → Inside cut, full depth (19mm) using 6mm bit (1/4” per pass)

Depth per pass

All cuts use 6–7 mm per pass for a total of 3 passes through 19 mm MDF. MDF's resin binder generates a lot of heat, so keeping passes shallow protects the bit and prevents the material from swelling or fuzzing around the hole walls. The 8 mm flat bit can handle slightly more but staying consistent across all cuts keeps things simple.

Cut path for holes

When cutting holes with Shaper Origin you want to use two separate operations, pocket and inside cut.

Pocket operation: Clears material from the *interior* of a shape by raster-filling it (the blue hash pattern you see). The Origin keeps the bit *inside* the boundary and doesn't cut right to the edge. It intentionally leaves a small ring of material between the pocket fill and the path line. If you look at the hole on your origin you will see that the blue hatch marks for the pocket operation doesn't reach the green outline of the hole. The result is a flat-bottomed cleared area but the walls are slightly undersized.

Inside cut operation: Runs the bit *along* the path line with the bit offset to the inside, cutting the wall of the hole to the exact diameter. This is what gives you the precise final diameter.

What you need to do:

The correct two-step process for a clean through-hole is exactly what you almost did but in the wrong order conceptually:

1. **Pocket** to clear the bulk of the material (which you've done)
2. **Inside cut** on the same circle to trim the wall to the exact 20 mm diameter

If your holes have only been cleared with a pocket cut then you have the wrong diameter. To fix this go back to that hole, select the circle, switch the operation to **Inside cut**, set depth to full depth of cut (19 mm for 3/4” MDF for example), and run it. The bit will trace the perimeter of the 20 mm circle and bring the hole to exact size. The material removal will be minimal — just that remaining ring — so it should cut cleanly and quickly.

Note that this process eliminates the need for leaving a 0.02 mm offset. This is a good thing to do for the outline of the shelf. Cut in 3 6mm passes with a 0.02mm offset then do a final pull depth pass.

The final profile pass

This is where you want to leave a thin onion-skin (0.2–0.3 mm) on the last pass, break the tabs, then flush-trim or sand smooth. This prevents the cutout piece from shifting and being grabbed by the bit at the last moment. Alternatively I used double sided tape and a sacrificial board under the work piece to use the 1/4” bit to but the final pass using the Shaper Origin.

The chamfer

The router edges of the shelf DO NOT get a chamfers. The 20mm holes do get a 4mm chamfer. I used the **Freud 40-104 chamfer bit** that has a 45° angle, ½″ bearing diameter (12.7mm), ½″ carbide height.

Note that the chamfer size is fixed by geometry of the bit, not by bit height. With a top-bearing chamfer bit, the bearing controls the cut, not the height setting. The bearing (radius 6.35mm) rides on the top face of the workpiece around the hole (radius 10mm). The difference between hole radius and bearing radius is what determines the chamfer:

- Chamfer depth (vertical and horizontal) = 10.0 − 6.35 = **3.65mm**
- Chamfer face width = **5.16mm**

So this bit on the 20mm hole will produce a **3.65mm chamfer**, not exactly 4mm — but that's very close and in practice indistinguishable.

Height setting:

The bit height doesn't control chamfer size here — the bearing does. Set the height so the **bearing bottom sits level with the top face of your workpiece** as it lies flat on the router table. Practically:

1. Place your shelf flat on the router table over the bit
2. Raise the bit until the bearing just contacts the top face of the shelf
3. Lock the height — you're done

The cutter will automatically engage the hole edge at the correct geometry. There's no arithmetic needed for the height — just get the bearing touching the top surface.

One caution: lower the bit fully before placing the workpiece over it, then raise slowly — don't lower the workpiece onto a spinning bit.

Large Shelf — Standard Variation

The standard large shelf top is the design produced and sold by Shaper Tools as part of their **Shelf Upgrade Kit** (SKU: SU1-WSUK1). It is a plain, unperforated 18mm MDF top measuring 410 × 280mm (16⅛" × 11") — over 3× bigger than the Shelf Assembly included with the base Workstation — and is the starting point from which the Vacudog and MFT variations are derived. Shaper Tools

The shelf slides onto the Workstation's clamping face and locks at the desired height via a lever. An adjustable support leg keeps the surface flat and coplanar with the top of the Workstation under load, with both coarse and fine-tune height adjustment spanning a range of 29⅛" to 44⅞" — accommodating a wide range of bench heights. Shaper Tools

The purpose of the large shelf is straightforward: it provides a flat, rigid, horizontal work surface that extends the effective cutting area of the Workstation significantly. Whether you're driving Origin manually or using it with BenchPilot, the Shelf Upgrade Kit gets you set up quickly and provides a stable platform for horizontal cuts. Typical uses include routing pockets and profiles into the face of a workpiece, sanding, and any operation where the material needs to lie flat rather than stand vertically against the Workstation's clamping face. Shaper Tools

The shelf top itself is a consumable — it is designed to be replaced as it becomes worn or damaged from repeated use as a spoilboard, and Shaper publishes the cut file (the SVG provided here) so that users can mill their own replacements from 18mm MDF using their Shaper Origin. The eight bolt holes accept M6 × 12mm fasteners to attach the top to the aluminium shelf body, with counterbored pockets that keep the bolt heads flush with or below the surface so they never interfere with workpieces or the Origin's base.

The key dimensions are listed below and augment what can be seen in the official dimensions provided by Shaper shown in figure 1.

- Counterbores: Ø12.7mm (r=6.35), 12mm deep
- Through holes: Ø6.5mm (r=3.25), 18mm through
- Tab: 290mm wide (x=60–350), 15mm deep

- Board body: x=10–400, y=15–280
- **0 groups, 0 open paths —** this is relevant for defining cut paths in Shaper studio.

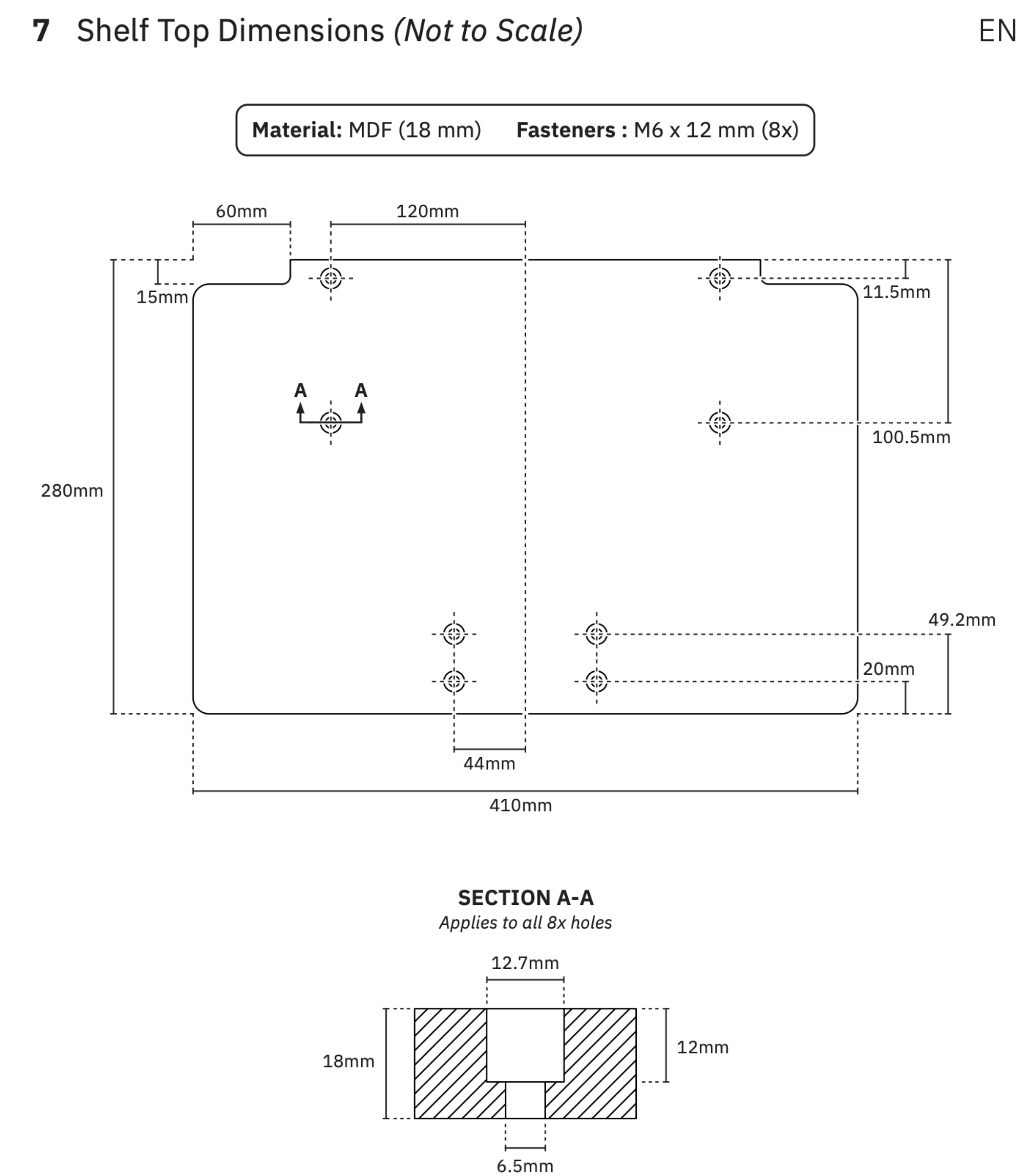

Figure 1: Shaper Large Shelf official Dimensions

Large Shelf — Vacudog Variation

The Vacudog variation of the Shaper Workstation Large Shelf is designed for clamp-free workholding using vacuum suction. It differs from the standard shelf only in the addition of a single Ø20mm hole routed through the centre of the board, which serves as the connection point for a Vacudog vacuum fitting.

The Vacudog is a compact, patent-pending vacuum fitting adapter made by Bench Dogs (UK) that presses into any standard 20mm bench dog hole. Once inserted, a rubber seal is placed around it to create a tight perimeter, and a vacuum pump is connected via an 8mm push-to-connect tube fitting run through or under the shelf. When the pump is switched on, the workpiece is held flat against the shelf surface by atmospheric pressure — no clamps required. The flush, non-protruding design allows the Vacudog to live in the bench full-time without interfering with normal work, and the vacuum line routes below the surface, eliminating hoses and clamps that would otherwise obstruct the Shaper Origin as it moves across the material.

This is particularly useful when routing or sanding smaller workpieces on the Shaper Workstation, where traditional clamps can interfere with Origin's travel path. The single central hole keeps the design simple — one Vacudog unit is sufficient for most workpiece sizes, with the rubber seal sized to suit the material being held.

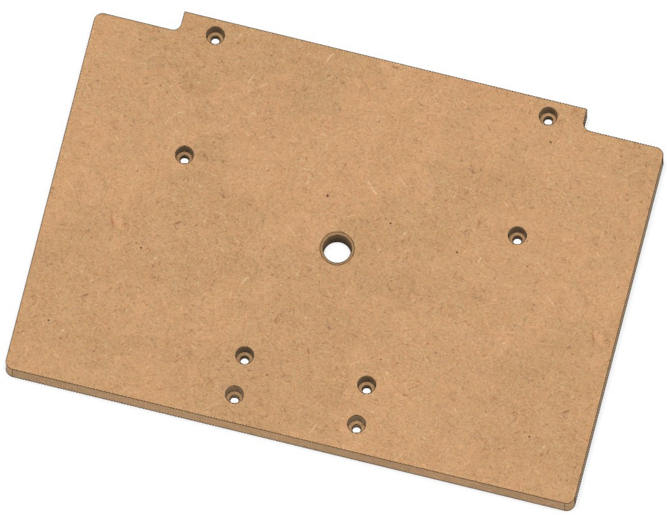

Figure 2: Rendering of Vacudog large shelf variatrion

Large Shelf — MFT Variation

The MFT variation of the Shaper Workstation Large Shelf replaces the single central hole with a full grid of Ø20mm dog holes, turning the shelf into a miniature MFT-style work surface compatible with the broad ecosystem of MFT clamping accessories.

MFT stands for **Multi-Function Table**, a workbench concept originated and popularised by Festool. The defining feature of an MFT is a top surface drilled with a regular grid of 20mm holes spaced 96mm apart centre-to-centre — a spacing derived from European cabinet-making standards. This uniform hole pattern provides the basis for secure and repeatable clamping using bench dogs, Festool-compatible clamps, and a wide range of third-party accessories, making the surface suitable for routing, sanding, sawing, and assembly work. The format has become an open standard adopted well beyond Festool's own product line, with many manufacturers producing compatible dogs, stops, squares, and vacuum accessories.

The MFT shelf uses different hole spacings from that of a standard MFT grid. This different spacing of the grid was adapted to fit the 410×280mm shelf, with 7 columns and 5 rows of holes (33 holes total, with 2 omitted where they would conflict with the mounting bolt positions). Since we still use 20mm holes, any MFT-compatible bench dog, hold-down clamp, or Vacudog vacuum fitting to be used anywhere across the shelf surface, giving maximum flexibility for positioning and holding odd-shaped or small workpieces directly on the Workstation.

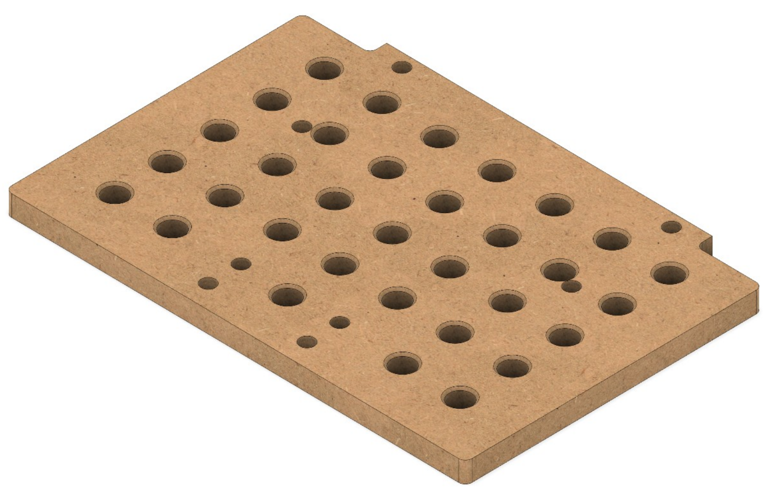

Figure 3: MFT large shelf variartion